UCLouvain

École polytechnique de Louvain

ÉCOLE POLYTECHNIQUE DE LOUVAIN

LEPL2022 - Health Innovation Classes

Empathizing - Hackathon

February 3rd, 2026

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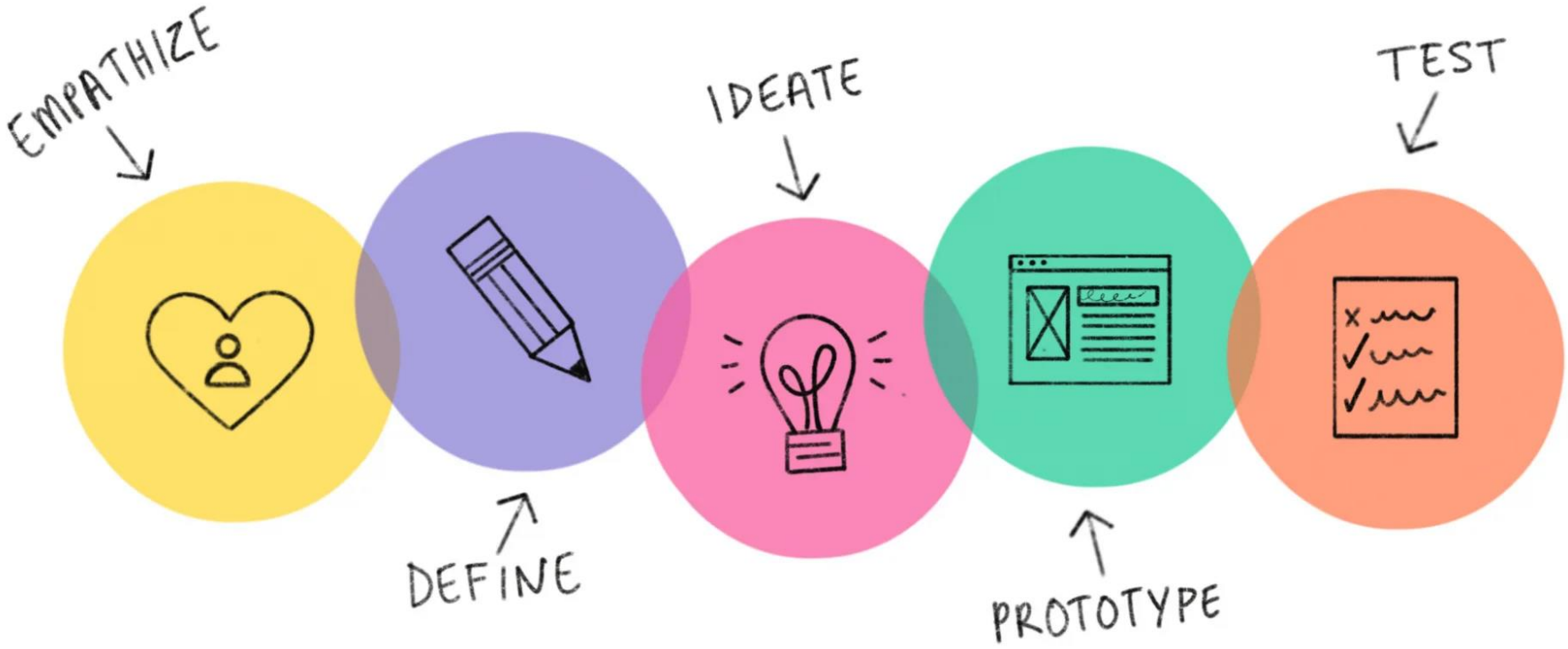
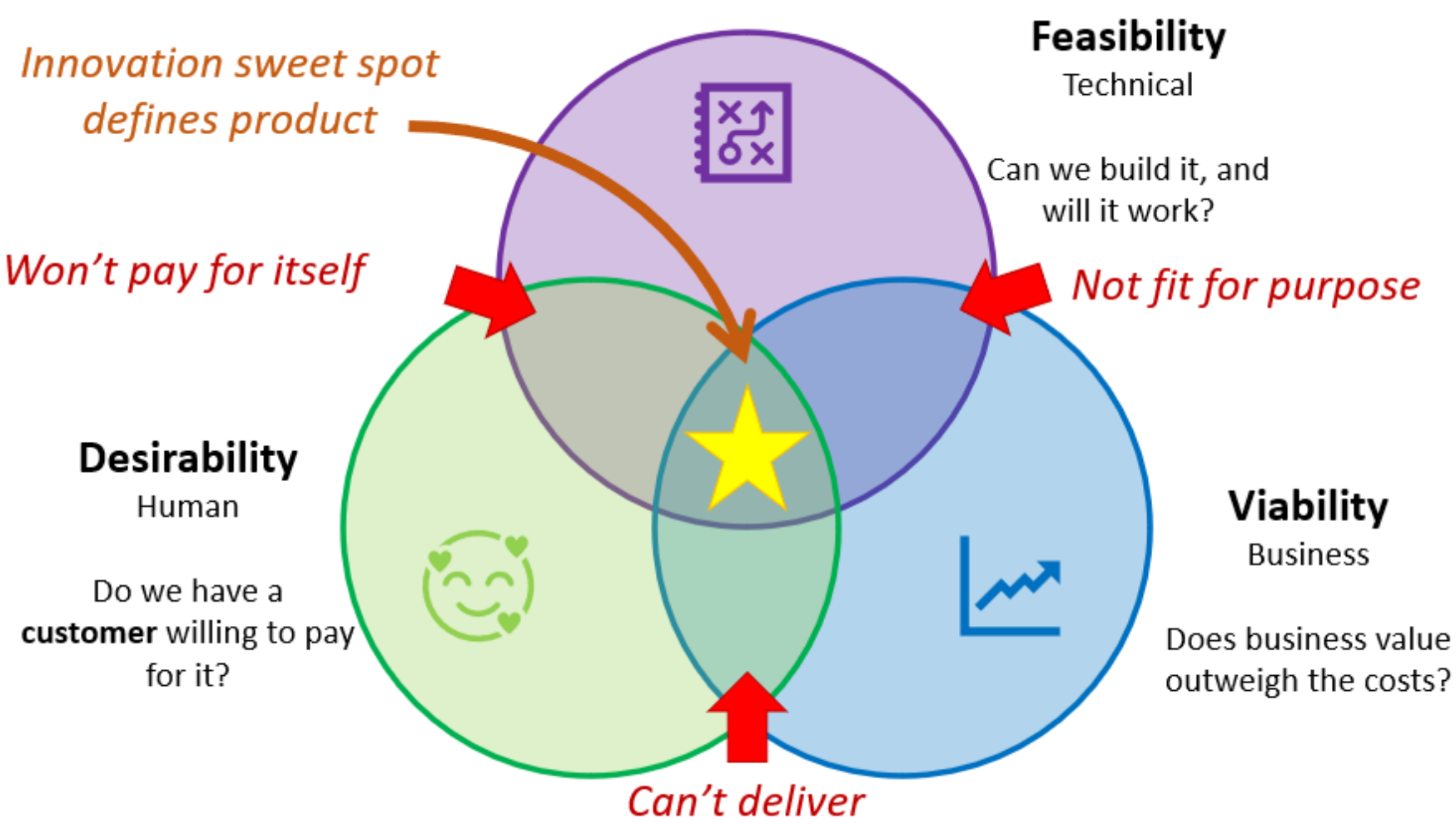
Reminder

DESIGN THINKING PROCESS




Innovation sweet spot defines product
Feasibility
Technical
Can we build it, and will it work?
Won't pay for itself
Not fit for purpose
Desirability
Human
Do we have a **customer** willing to pay for it?
Viability
Business
Does business value outweigh the costs?
Can't deliver

Empathizing

- Google Glass was promoted for surgeons and clinicians to access patient data hands-free, record procedures, or receive real-time guidance.

Why it failed (empathy gap):

- **Ignored clinician workflow:** Doctors found it distracting rather than helpful during patient interactions.
- **Patient discomfort:** Patients felt uneasy or surveilled when clinicians wore a camera on their face.
- **Social and ethical blind spots:** Privacy concerns and trust were underestimated.
- **Cognitive overload:** The device added information at moments when clinicians needed focus, not more inputs.

EXERCISE

4 different interview targets

Busy doctor

Nurse on ward

Hospital pharmacist

Chronic patient

INDIVIDUALLY

"When I interview a ___, they will probably say that their main problem is ___."

INDIVIDUALLY

- What this person *cares most about*
- What frustrates them daily
- What they expect from an innovation or solution

INDIVIDUALLY

Based on this, create 3 questions you would ask

COMPARE

“How many of you had *similar answers*?”

“How many of you feel quite confident you’re right?”

You have just designed a **fictional persona made of your assumptions**

and…

your **questions** were designed to **confirm your model** instead of testing it.

REWRITE ONE QUESTION OF YOUR CHOICE

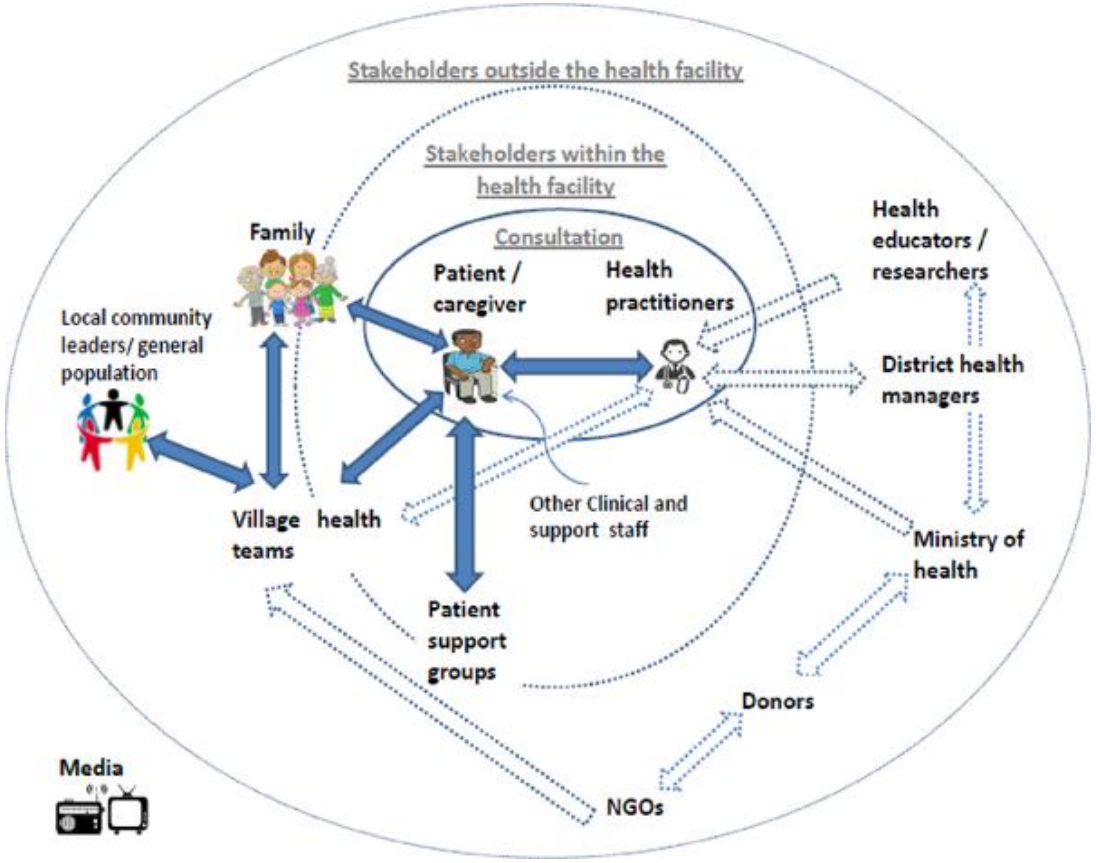
- (In)validate your assumptions
- Invite stories, not opinions
- Surface contradiction
- Make their real needs emerge

EMPATHIZING IS THE FOUNDATION OF ALL SUBSEQUENT STEPS

SEQUENCE

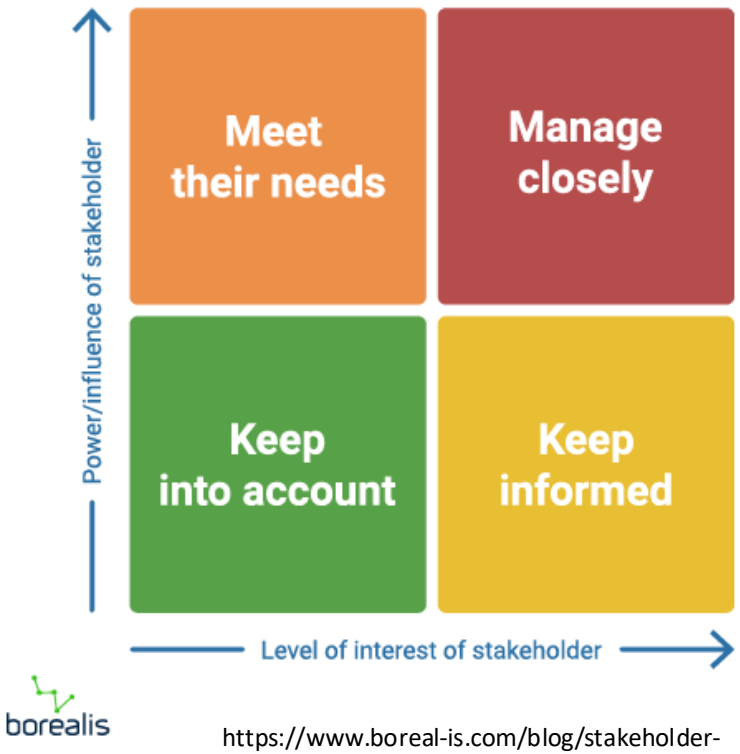
1. Identify stakeholders
2. Map stakeholders
3. Assess their respective influence and influence on the project
4. Select the most important
5. Create personas
6. Empathize
7. Create empathy maps

IDENTIFY AND MAP STAKEHOLDERS

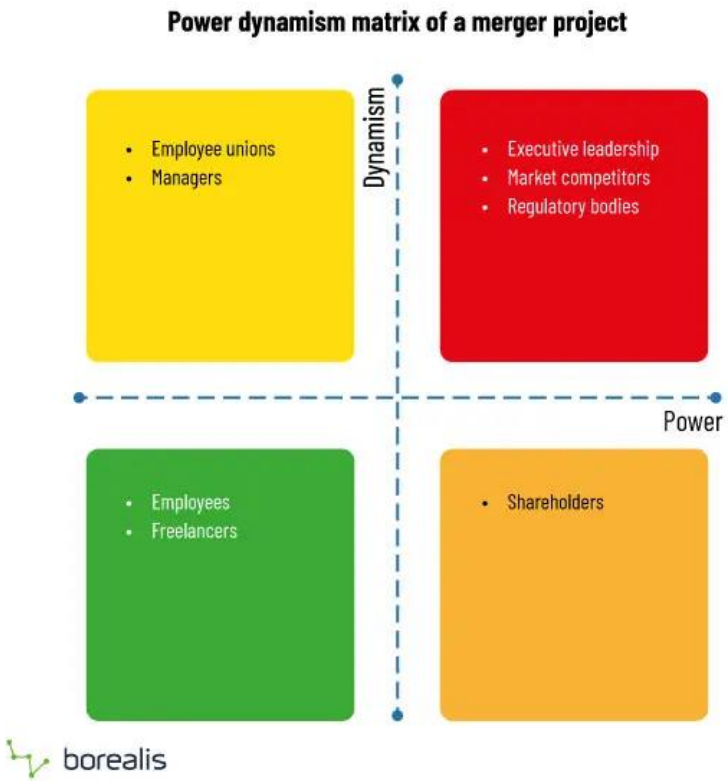


Waweru et al. - Stakeholder perceptions on patient-centered care at primary health care level in rural eastern Uganda: A qualitative inquiry

ASSESS THEIR INFLUENCE (some examples)



https://www.boreal-is.com/blog/stakeholder-mapping-identify-stakeholders/



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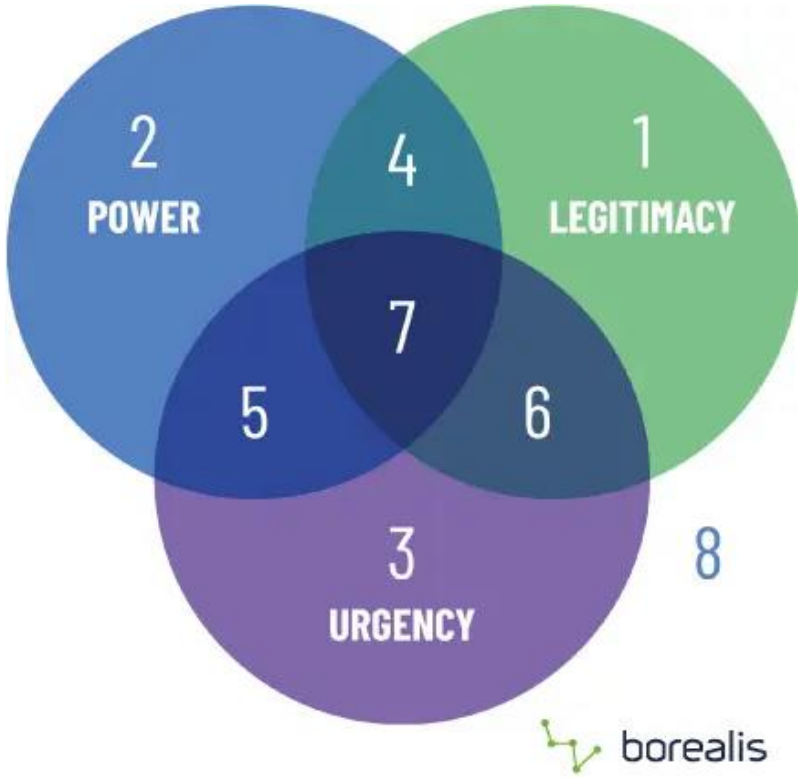

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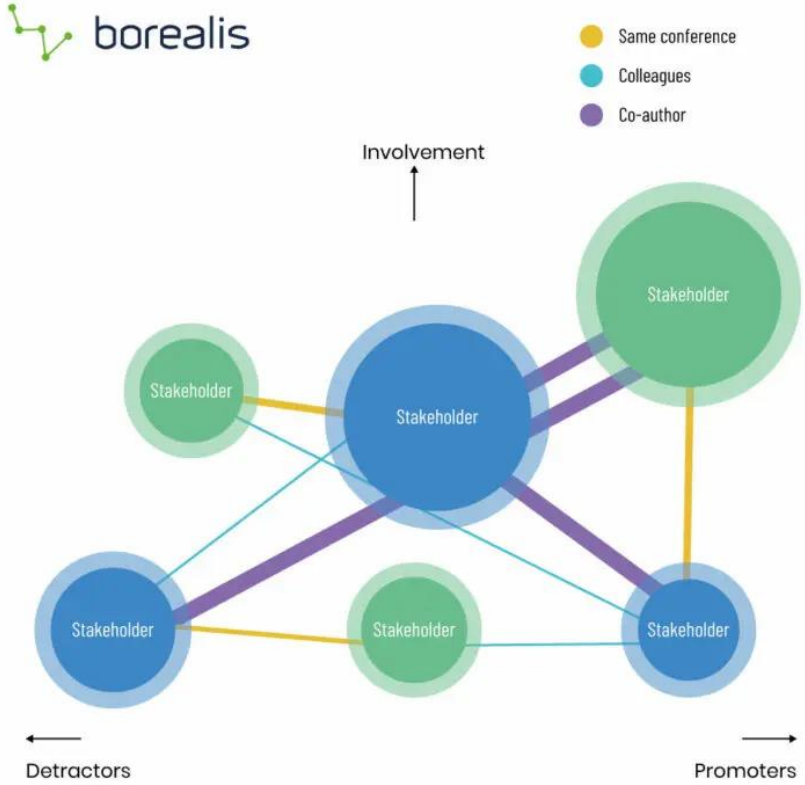

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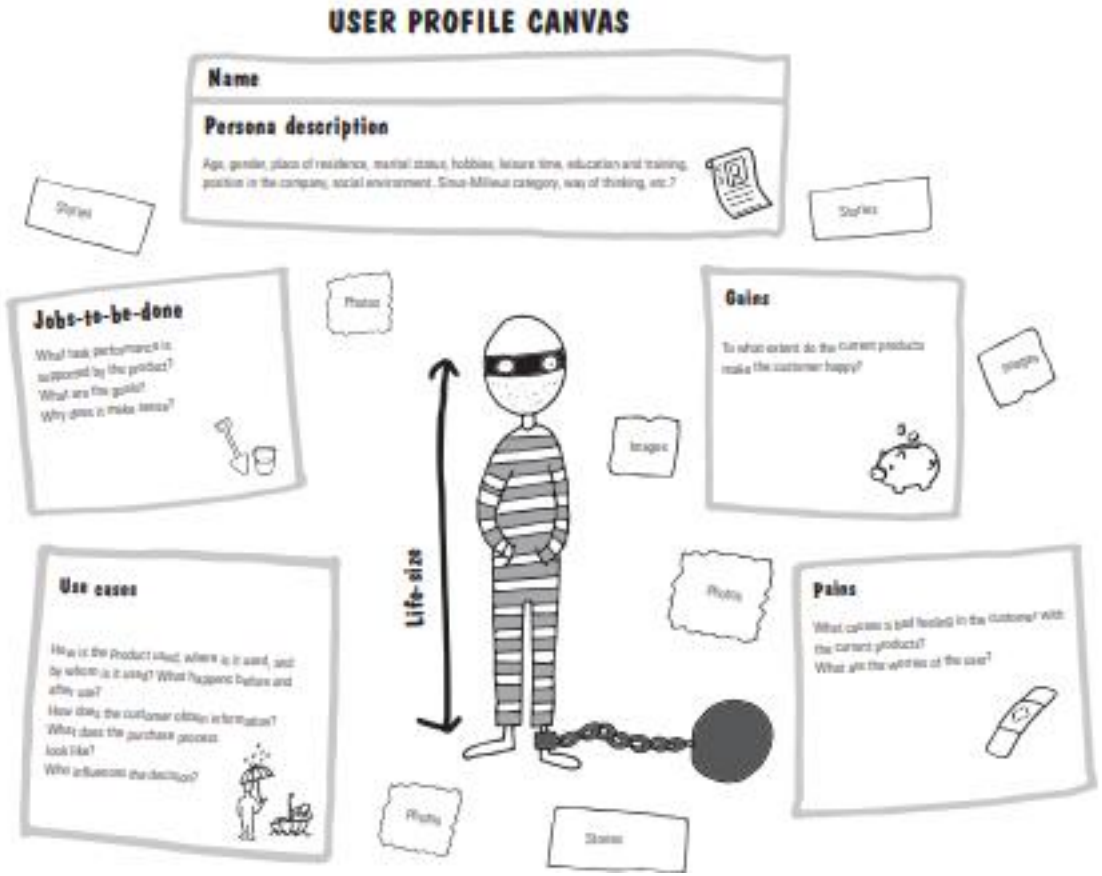
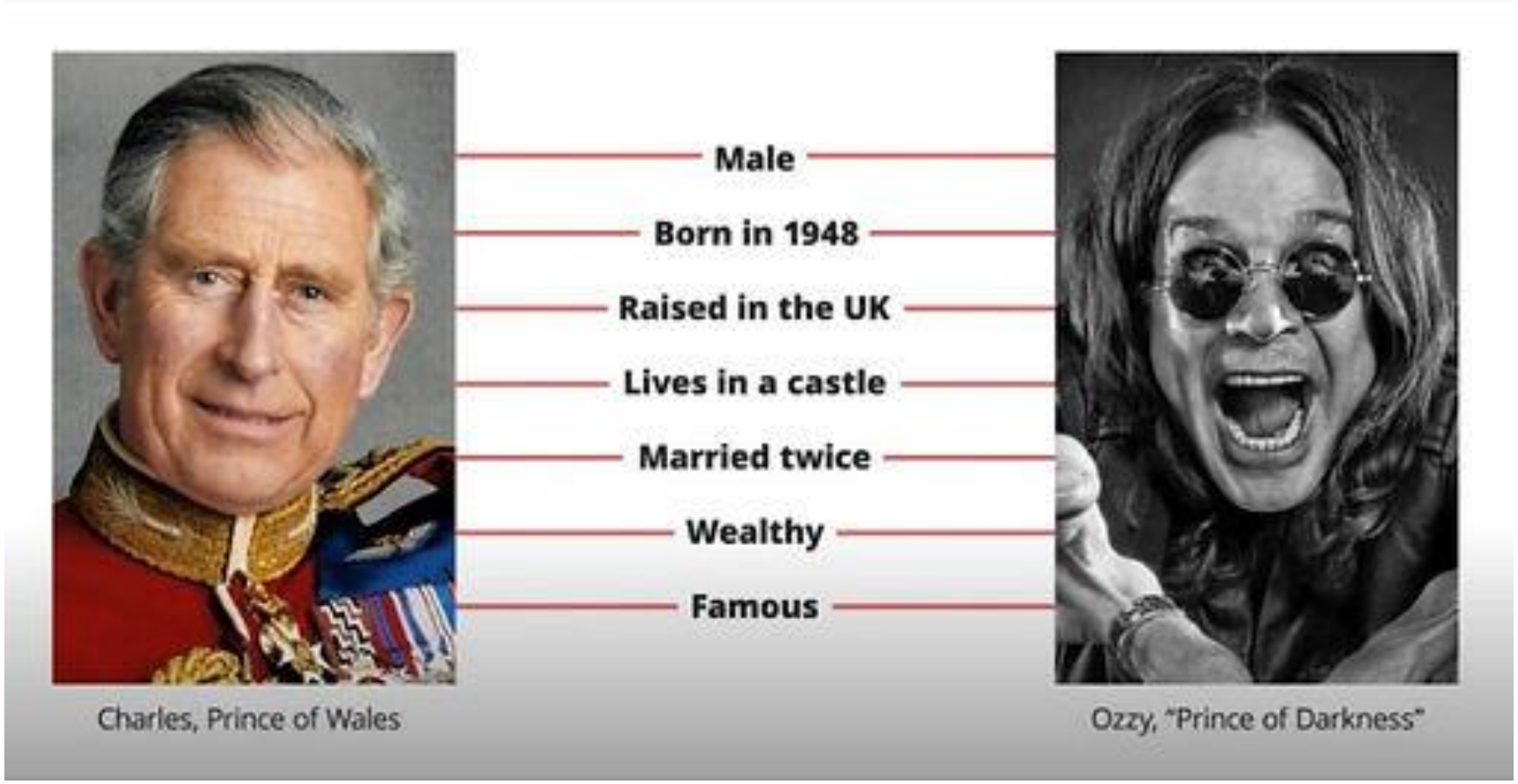
CREATE PERSONA



PERSONA : NOT ONLY ABOUT DEMOGRAPHICS



EMPATHIZE - SOURCES

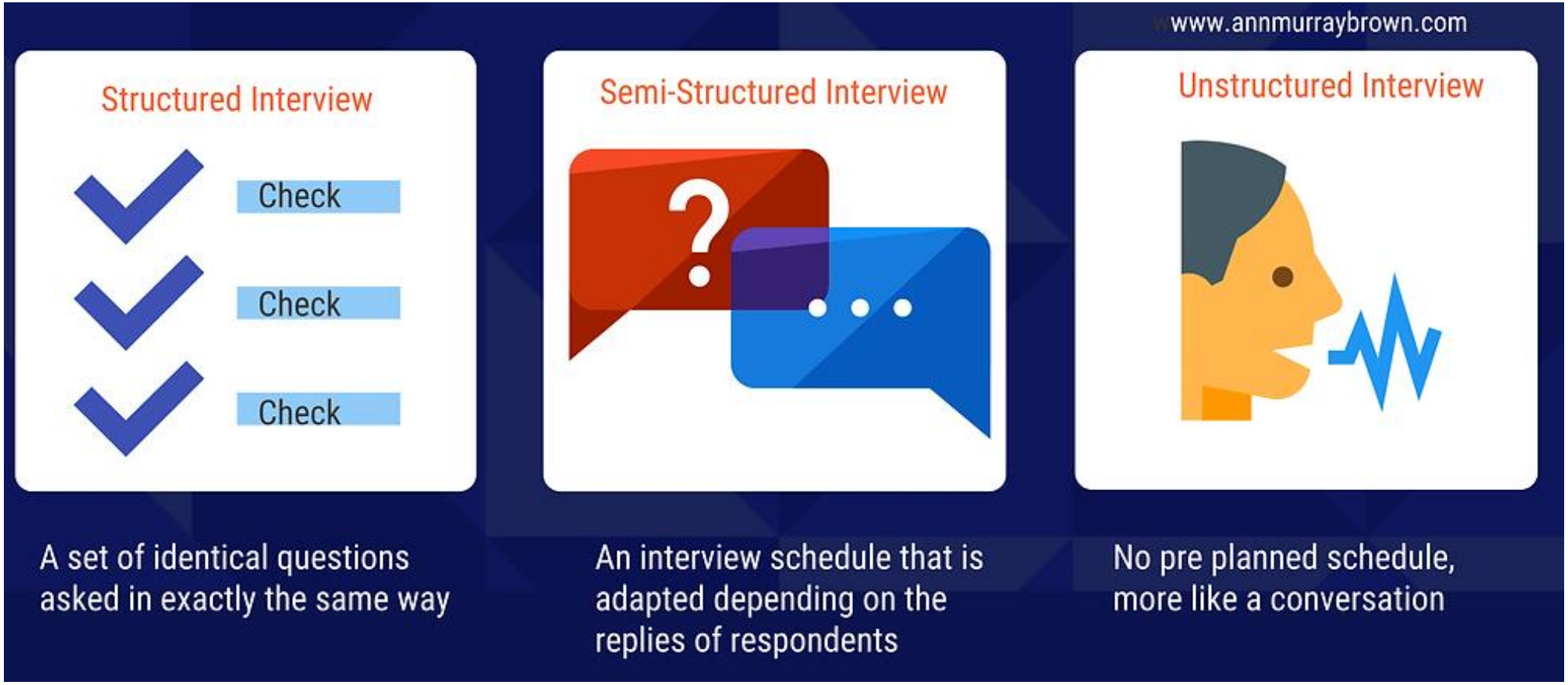
- Primary sources
  - User interviews
  - Field observations / shadowing
  - Surveys
- Secondary sources
  - Scientific literature
  - Reports, statistics, benchmarks
  - Existing user feedback / complaints
- Contextual data
  - Environment, constraints, workflows
  - Social, cultural, institutional factors

EMPATHIZE – RESEARCH METHODS

QUANTITATIVE		QUALITATIVE
Numerical and measurable data	DATA TYPE	Descriptive and non-numerical data (words, images, observations)
Objective and structured measurement	APPROACH	Subjective understanding and interpretation
Large and representative	SAMPLE SIZE	Small, purposefully selected
• How many? • How much? • How often?	RESEARCH QUESTIONS	• Why? • How? • What does it mean?
Surveys, experiments, and closed-ended questionnaires	DATA COLLECTION	Interviews, case studies, open-ended surveys and observations
Statistical tools: averages, percentages, regression, etc.	DATA ANALYSIS	Thematic or narrative analysis, coding, grounded theory
Generalisable results and predictive insights	RESULTS	Deep, contextual understanding and rich detail

https://schoolofleadership.co.uk/quantitative-vs-qualitative-research

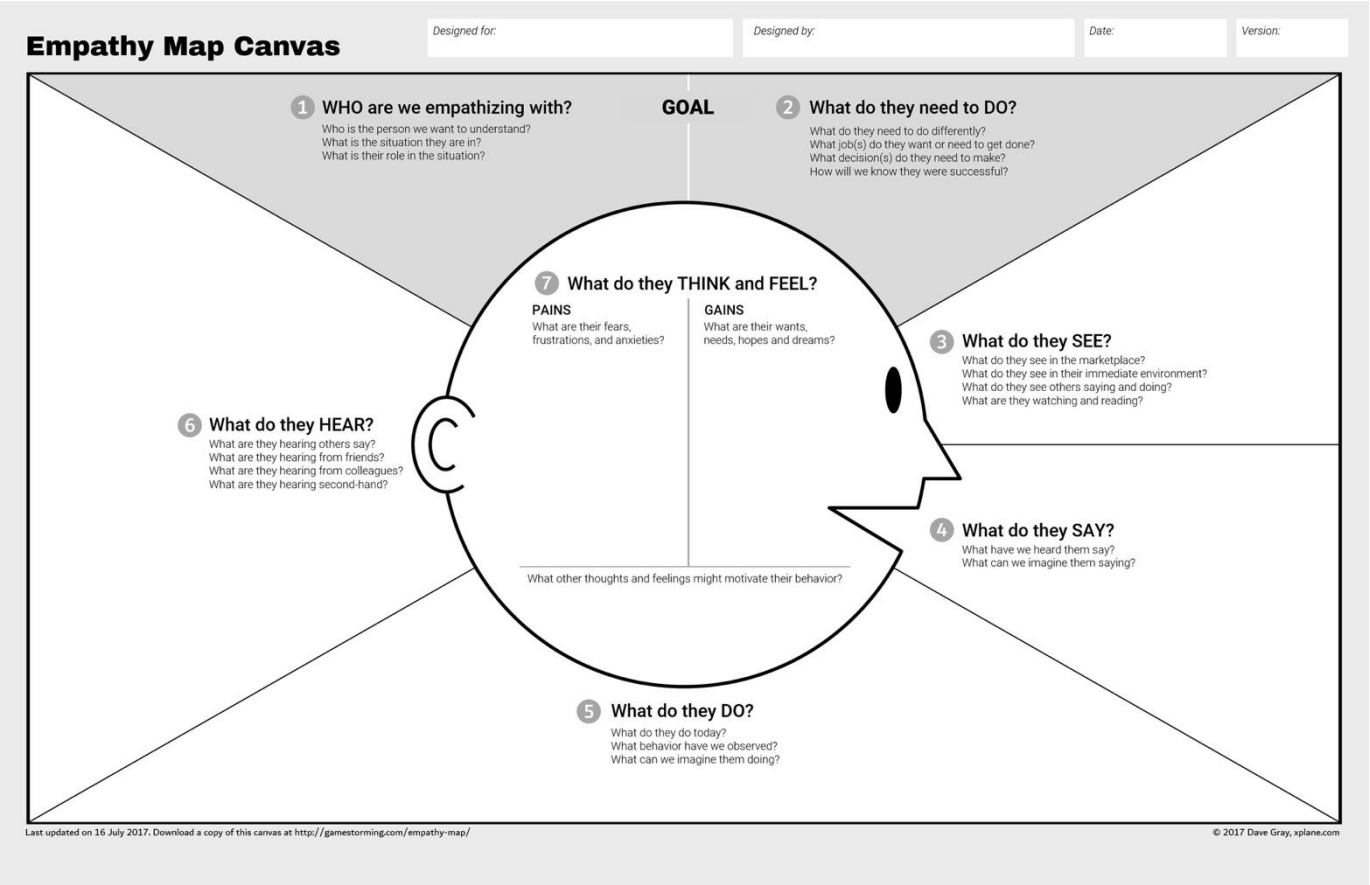
EMPATHIZE – INTERVIEW METHODS



https://www.annmurraybrown.com/single-post/types-of-interviews

NON-STRUCTURED	SEMI-STRUCTURED	STRUCTURED
Characteristics • No fixed interview guide • Conversation-led, user-driven • High flexibility, high depth • Harder to compare or analyze systematically • Strong dependence on interviewer skill **Typical Questions** • "Tell me about a typical day." • "What does this situation mean for you?" • "Can you tell me more about that?" Mostly narrative prompts and follow-ups **Best Used In Design Thinking** **Stage:** Early Empathize **Situation:** • When the problem space is unclear • When exploring sensitive or complex contexts • When users are experts of their own experience	**Characteristics** • Interview guide with key themes • Flexible order and wording • Allows follow-up questions and probing • Balance between consistency and depth • Most common in design research **Typical Questions** • "Can you walk me through your last experience with…?" • "What was difficult or frustrating in that situation?" • "Why was that important to you?" Mix of open-ended and probing questions **Best Used In Design Thinking** **Stage:** Empathize / Define **Situation:** • Early problem exploration • When you want rich insights but still some structure • Small to medium sample sizes	**Characteristics** • Predefined questions, same order for all participants • Little to no deviation • Easy to compare answers across users • Lower interviewer bias, lower depth **Typical Questions** • "How often do you use this service?" • "On a scale from 1 to 5, how satisfied are you?" • "Which option do you use most frequently?" Closed or very specific questions **Best Used In Design Thinking** **Stage:** Test / Validate **Situation:** • When hypotheses are already defined • When you need comparability or metrics • When combining with quantitative analysis

EMPATHY MAP



An empathy map helps teams understand users by capturing what they **think, feel, see, hear, say, and do**, revealing pains and gains to design solutions that truly fit their needs.

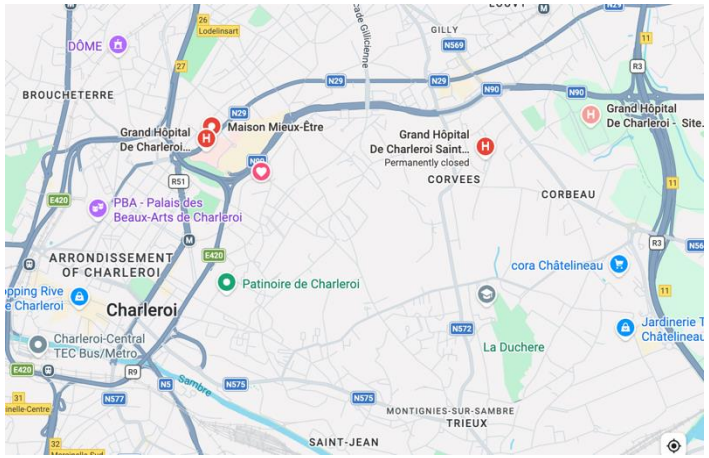
Hackathon

OBJECTIVES

- Go through the entire Design Thinking process in an accelerated manner
- Work on a concrete project. This year:

GHdC Central Pharmacy: Congestion and delays in handling returns of medicinal products

GRAND HOPITAL DE CHARLEROI



PLANNING

February 12th

- Arrival 8:30-8:45 at GHdC
- 9:00 start
- 9:00-9:45: presentation of the problem (incl. Field visit) + Q&A
- 9:45-10:00 Return to the room
- 10:00-12:30: Presentation and group work on empathy and problem definition
- 12:30-13:15: Lunch break (sandwich)
- 13:15-13:45: Another Q&A session
- 13:45-14:15: Finalizing the problem definition
- 14:15-16:30: Ideation and prototyping
- 16:30-17:30: Feedback from user representatives, presentation of the first iteration and prototype, feedback, Q&A
- 17:30-18:00 Wrap-up and closing of day 1, preparation for day 2

PLANNING

February 13th

- Arrival 8:30-8:45
- 9:00 Start
- 9:00-11:30: Work on the second iteration and associated prototype following feedback from the end of day 1
- 11:30-12:30: Final pitches
- 12:30: Lunch and departure

LOGISTICS

- Bus leaving at 8:00am **sharp** on Thursday from LLN
  Meeting point: https://maps.app.goo.gl/dfDNvxo7hb6b6WKB9
- Bus leaving from GHdC to LLN at 1:00pm on Friday
- Spend the night at Novotel Charleroi https://share.google/4bjPT6uZZ1Hqkqp3Y
  Evening dinner and breakfast included
- All travel between GHdC and city center by public transportation (TEC)

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